PrestaShop Deployment on AWS

Preliminary Assignment for Cyber Security Intern

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Implementation Documentation

Live Store URL:

http://51.21.182.10

Bincom Dev Center

1. Overview

This document describes the step-by-step process used to deploy a working PrestaShop e-commerce installation on Amazon Web Services (AWS), using only Free-Tier eligible resources. In line with the assignment requirements, the database is hosted on a separate server from the web application, and the final installation is reachable via a publicly accessible URL.

Architecture summary:

- Web/Application Server: Amazon EC2 instance (Ubuntu 26.04 LTS, t3.micro, Free Tier eligible)
- Database Server: Amazon RDS for MySQL (db.t3.micro, Free Tier eligible), hosted separately from the EC2 instance
- Networking: Both resources placed in the same default VPC; a security group rule permits the EC2 instance to reach the RDS instance on port 3306, while the database itself remains non-publicly accessible
- Application: PrestaShop 9.1.5 (latest stable release at time of installation), installed via the official web-based installer

Public URL: http://51.21.182.10

2. Setting Up the Database Server (Amazon RDS)

The database was provisioned first, using Amazon RDS, so that it could be referenced during the EC2 setup and PrestaShop installation steps that follow. Hosting the database on RDS, rather than installing MySQL directly on the EC2 instance, satisfies the requirement that the database must not be hosted on the same server as the application.

2.1 Engine and Configuration Method

MySQL was selected as the database engine, since PrestaShop requires a MySQL-compatible database. 'Full configuration' was chosen (rather than 'Easy create') to allow explicit control over public accessibility, security groups, and other settings critical to this assignment's requirements. The Free Tier template was selected to remain within AWS Free Tier limits.

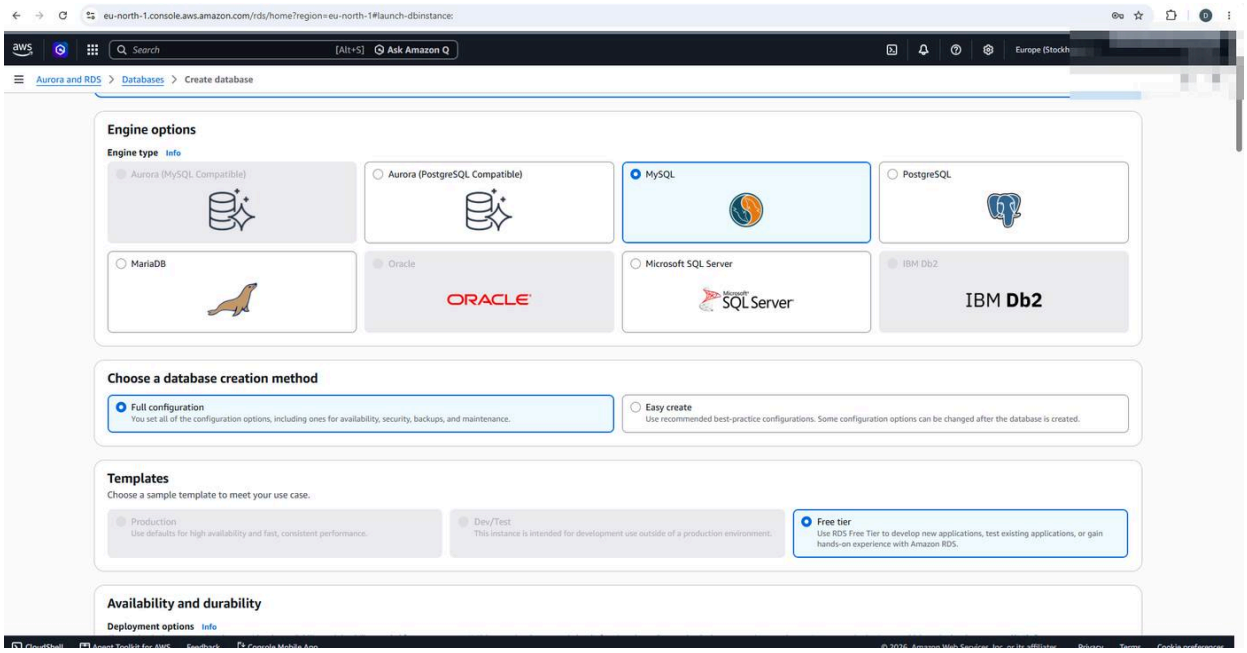


Figure 1: RDS engine type (MySQL), configuration method (Full configuration), and Free Tier template selected.

2.2 Instance Identification and Credentials

The database instance was named prestashop-db, with a master username of admin and a self-managed, strong password. A db.t3.micro instance class was used, which is Free Tier eligible.

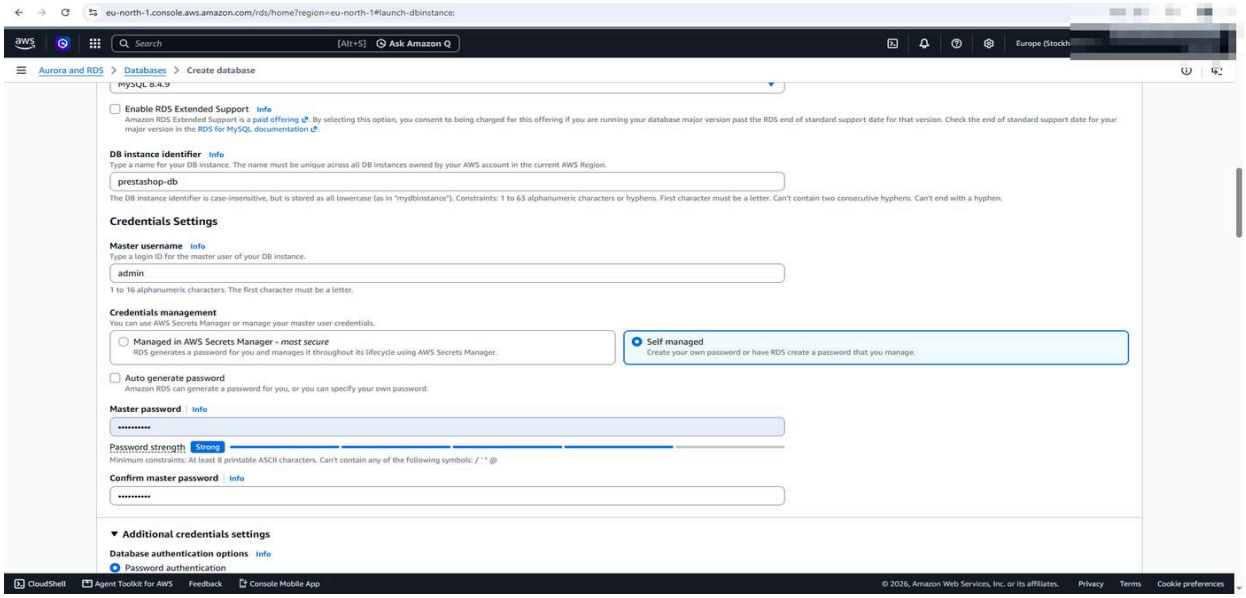


Figure 2: RDS instance identifier and master credentials configuration.

2.3 Connectivity and Public Access- Key Security Decision

This is one of the most important configuration decisions in the assignment. The database was placed in the same default VPC as the EC2 instance, but was explicitly NOT connected directly during creation, so the linking could be controlled manually via security groups.

Public access was set to No. This is deliberate: it ensures the database has no public IP address, and can only be reached from resources inside the VPC (specifically, the EC2 instance), not from the open internet. This directly satisfies both the database-isolation requirement and general security best practice.

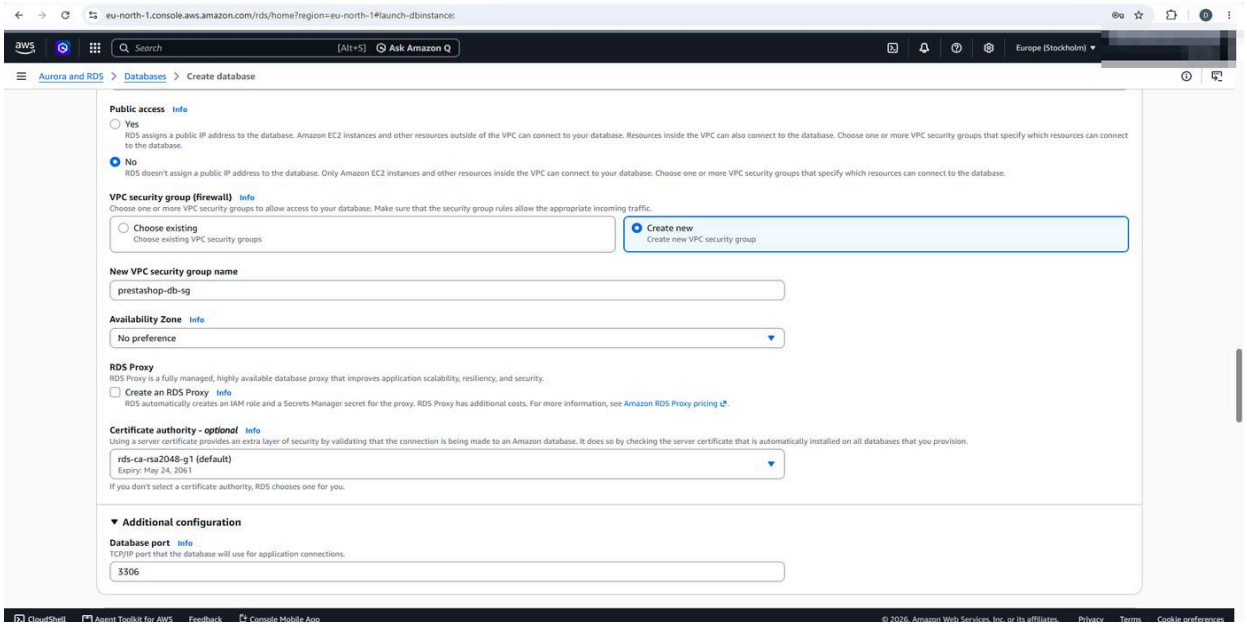


Figure 3: Connectivity settings — VPC, DB subnet group, and Public Access set to 'No'.

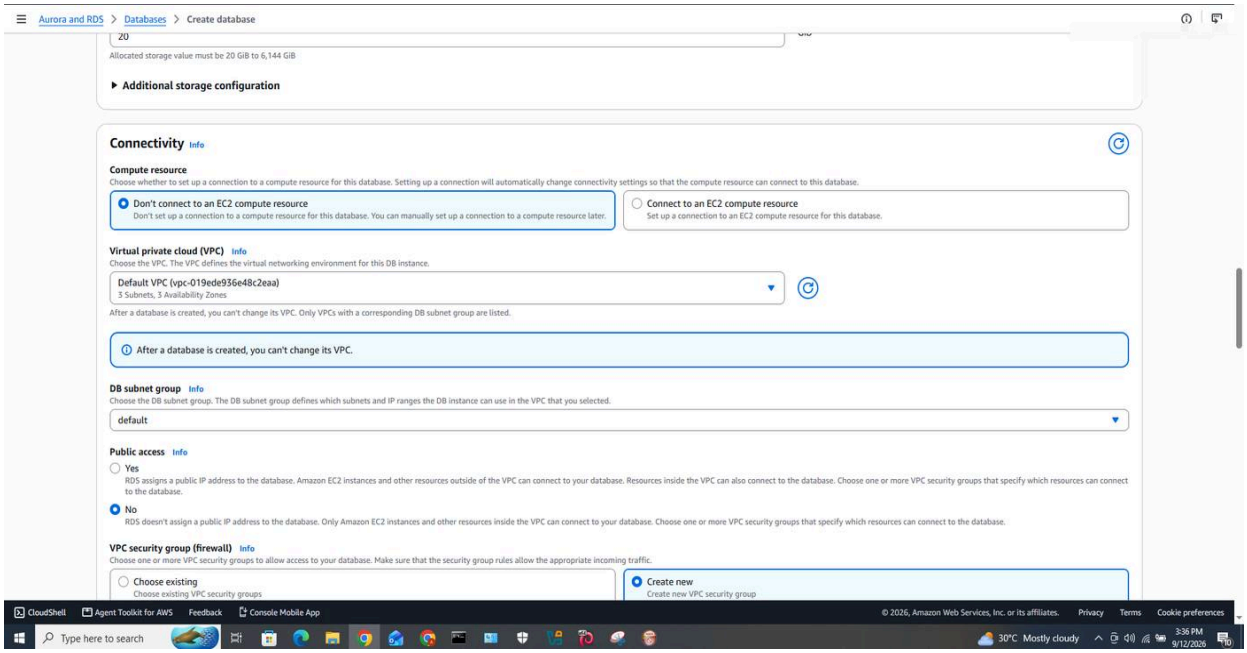


Figure 4: A new dedicated VPC security group (prestashop-db-sg) was created specifically for the database.

2.4 Database Name and Additional Configuration

An initial database name of prestashop was specified so that RDS automatically created the schema PrestaShop's installer would later connect to. Encryption and automated backups were left enabled (AWS defaults), which does not conflict with Free Tier usage.

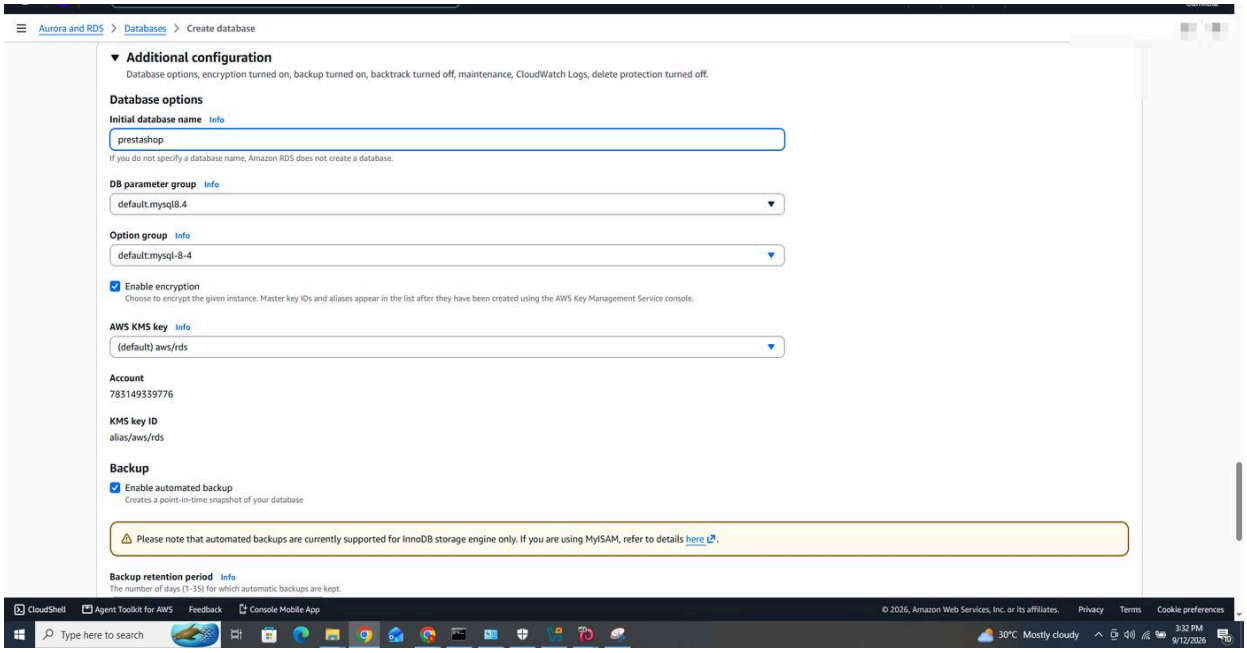


Figure 5: Initial database name set to 'prestashop', with encryption and automated backups enabled.

Once these settings were confirmed, the database was created and took several minutes to become 'Available' before the EC2 instance was configured.

3. Setting Up the Application Server (Amazon EC2)

With the database provisioning in the background, an EC2 instance was launched to host the PrestaShop application and web server.

3.1 Instance Configuration

- AMI: Ubuntu Server 26.04 LTS (Free Tier eligible)
- Instance type: t3.micro (Free Tier eligible)
- Key pair: newly generated RSA key pair (prestashop-key.pem) for SSH access
- Network: same default VPC as the RDS instance, to allow future connectivity

3.2 Security Group (Firewall) Rules

A new security group, prestashop-web-sg, was created for the EC2 instance with three inbound rules:

- SSH (port 22) - for remote terminal administration
- HTTP (port 80) - to allow public visitors to reach the PrestaShop storefront
- HTTPS (port 443) - reserved for secure traffic, should SSL be configured later

HTTP and HTTPS were deliberately opened to all sources (0.0.0.0/0), since the assignment requires the storefront to be publicly accessible. SSH access could be further restricted to a specific IP in a production environment.

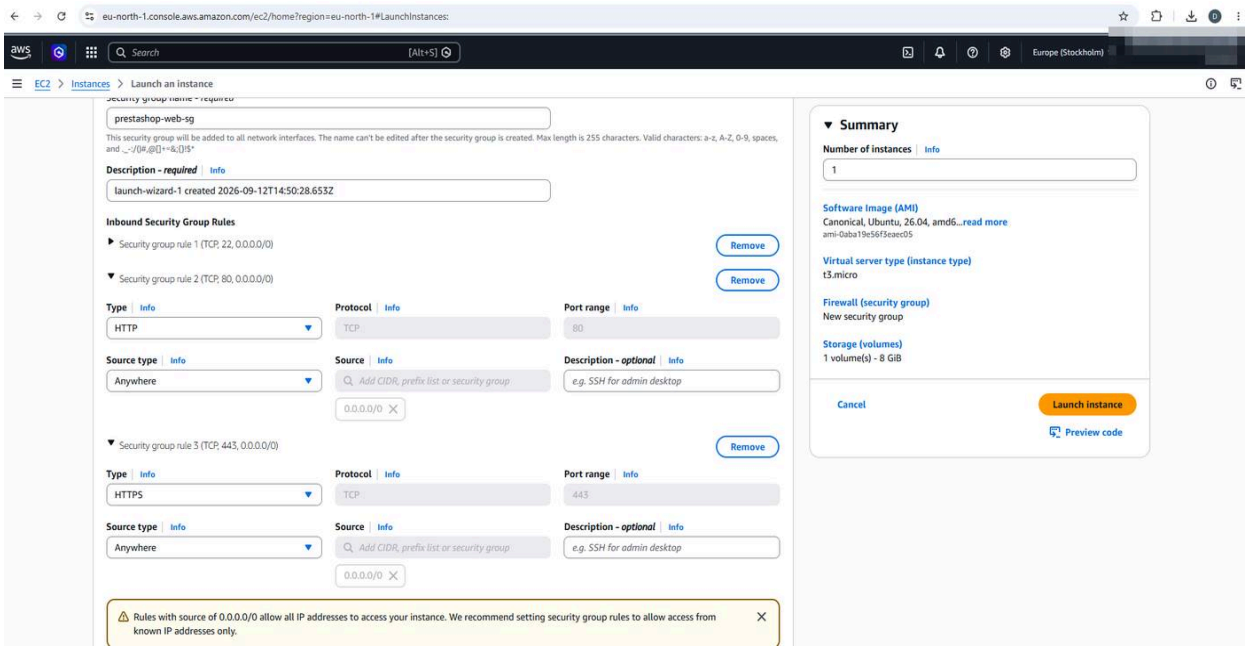


Figure 6: EC2 security group inbound rules - SSH, HTTP, and HTTPS.

3.3 Instance Launched

After launch, the instance reached the 'Running' state with a status check of 3/3 checks passed, and was assigned a public IPv4 address of 51.21.182.10, this address became the final public URL for the store.

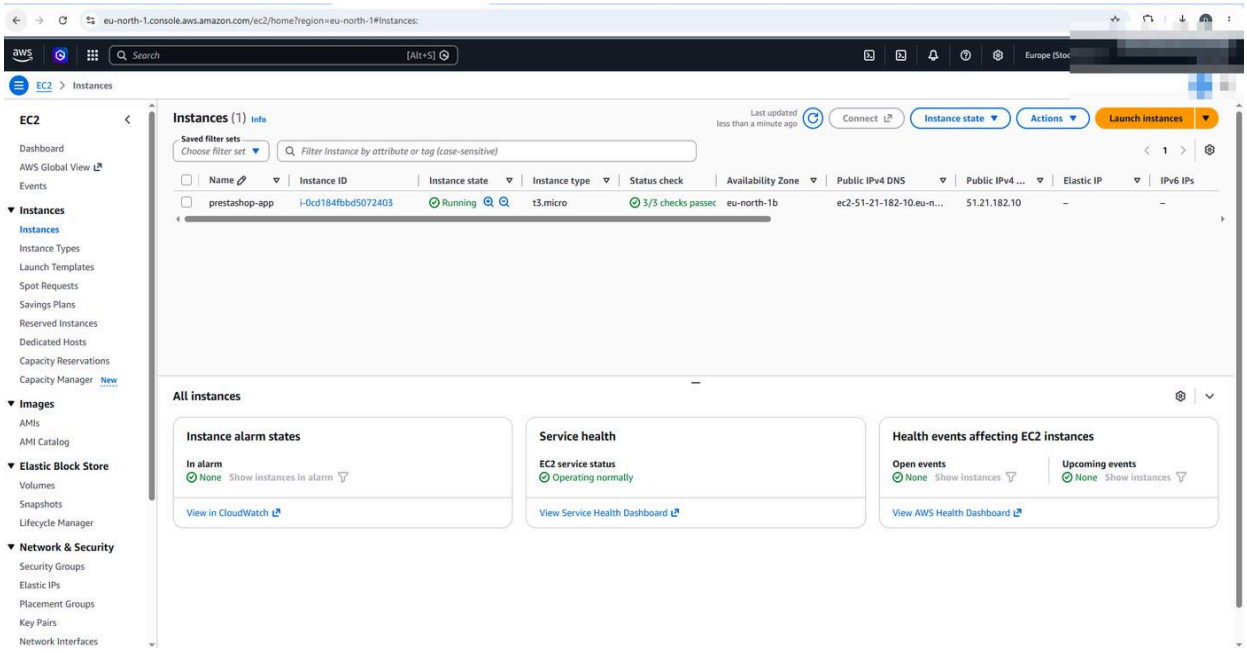


Figure 7: EC2 instance running successfully with public IP 51.21.182.10.

4. Linking the Application Server to the Database

Because the RDS instance was configured with no public access, an explicit rule was needed to allow only the EC2 instance, not the wider internet to reach the database on port 3306 (the default MySQL port).

This was done by editing the inbound rules of the RDS security group (prestashop-db-sg) to add a MySQL/Aurora rule whose source was set to the EC2 security group (prestashop-web-sg) itself, rather than an IP address or range. This means only traffic originating from resources using that specific security group (i.e., the EC2 instance) is permitted to reach the database, a tighter and more maintainable rule than allowing a fixed IP range.

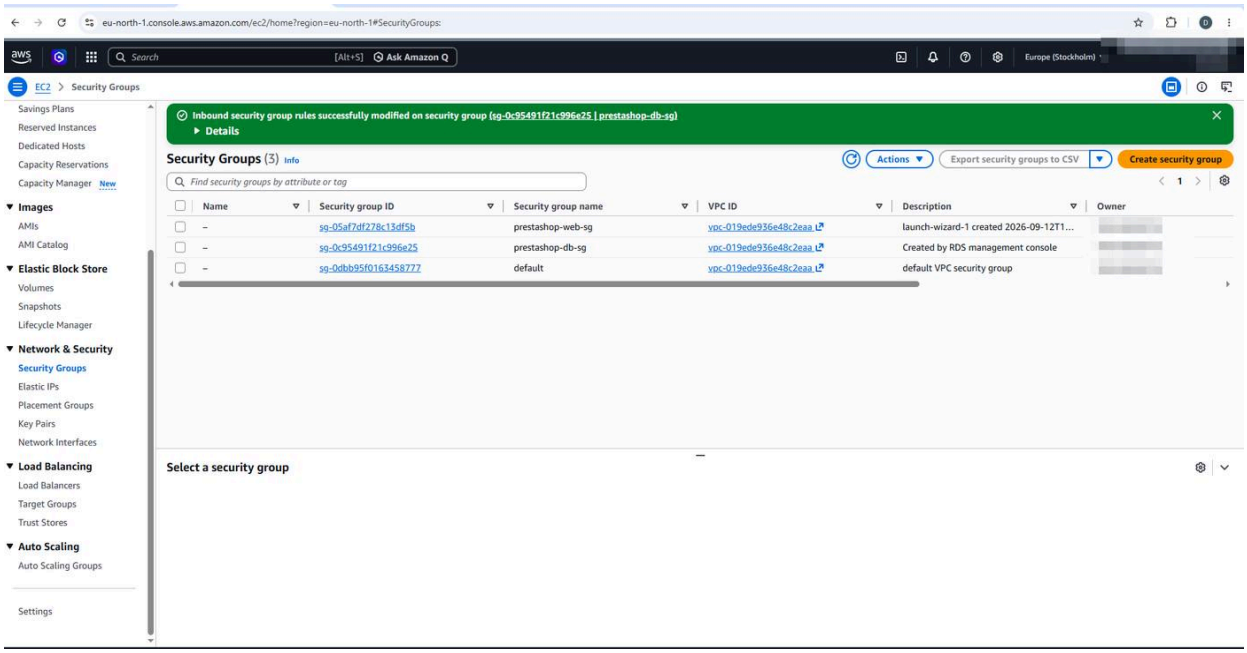


Figure 8: Confirmation banner showing the inbound rule was successfully added to the RDS security group, referencing the EC2 security group as the trusted source.

5. Connecting to the Server and Installing Software

The EC2 instance was accessed using AWS's browser-based EC2 Instance Connect feature, avoiding the need for a separate SSH client.

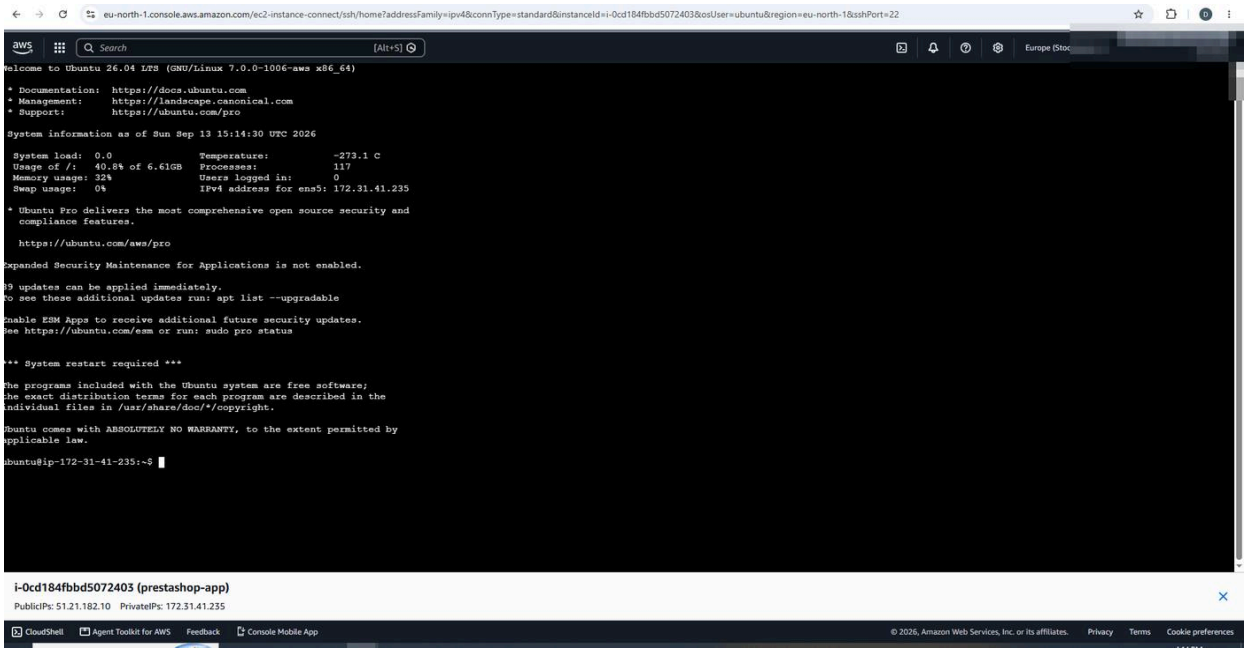


Figure 9: Successful SSH connection into the EC2 instance via EC2 Instance Connect.

5.1 Installing the Web Server Stack

The following software was installed on the instance via the terminal:

- Apache2 - web server
- PHP 8.5 and required extensions (mysql, gd, curl, zip, xml, mbstring, intl, bcmath)
- Unzip - for extracting the PrestaShop package

Apache was enabled to start automatically and confirmed as 'active (running)'.

5.2 Downloading PrestaShop

PrestaShop 9.1.5 (the latest stable release, chosen for its official support of PHP 8.5) was downloaded directly onto the server using wget, then extracted into the Apache web root at /var/www/html/prestashop. File ownership was set to the www-data user, and Apache's DocumentRoot and virtual host configuration were updated to point to this new folder.



Figure 10: PrestaShop 9.1.5 package (116 MB) downloaded successfully to the EC2 instance.

5.3 Server Tuning

During installation, two server-level adjustments were required to accommodate the small memory footprint of a t3.micro instance and current PHP behaviour:

- PHP execution and memory limits were raised (max_execution_time, memory_limit, upload_max_filesize, post_max_size) to prevent installer timeouts.
- A 1GB swap file was added, since the instance's ~900MB of RAM was insufficient on its own for the PrestaShop installer, which was intermittently being terminated by the Linux Out-Of-Memory killer during heavier steps (theme and module installation).
- Apache was configured with an explicit <Directory> block granting AllowOverride All for the PrestaShop folder, which was required for PrestaShop's .htaccess-based URL rewriting (and therefore the admin panel URL) to function correctly.

6. Running the PrestaShop Web Installer

With the server prepared, the public IP address was visited in a browser, launching PrestaShop's Installation Assistant.

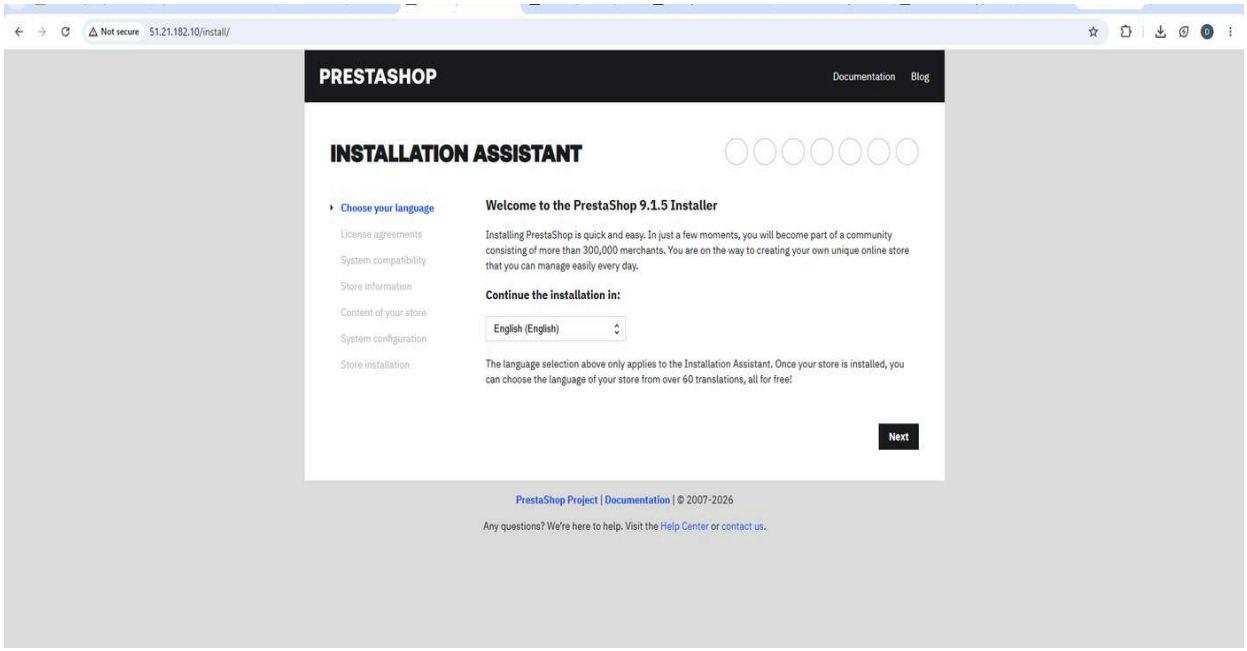


Figure 11: PrestaShop 9.1.5 Installation Assistant, reachable publicly at http://51.21.182.10.

6.1 Database Connection

At the System Configuration step, the installer was pointed at the RDS endpoint (rather than a local database), using the credentials configured in Section 2. A successful 'Database is connected' message confirmed that the EC2-to-RDS security group linking (Section 4) was working correctly.

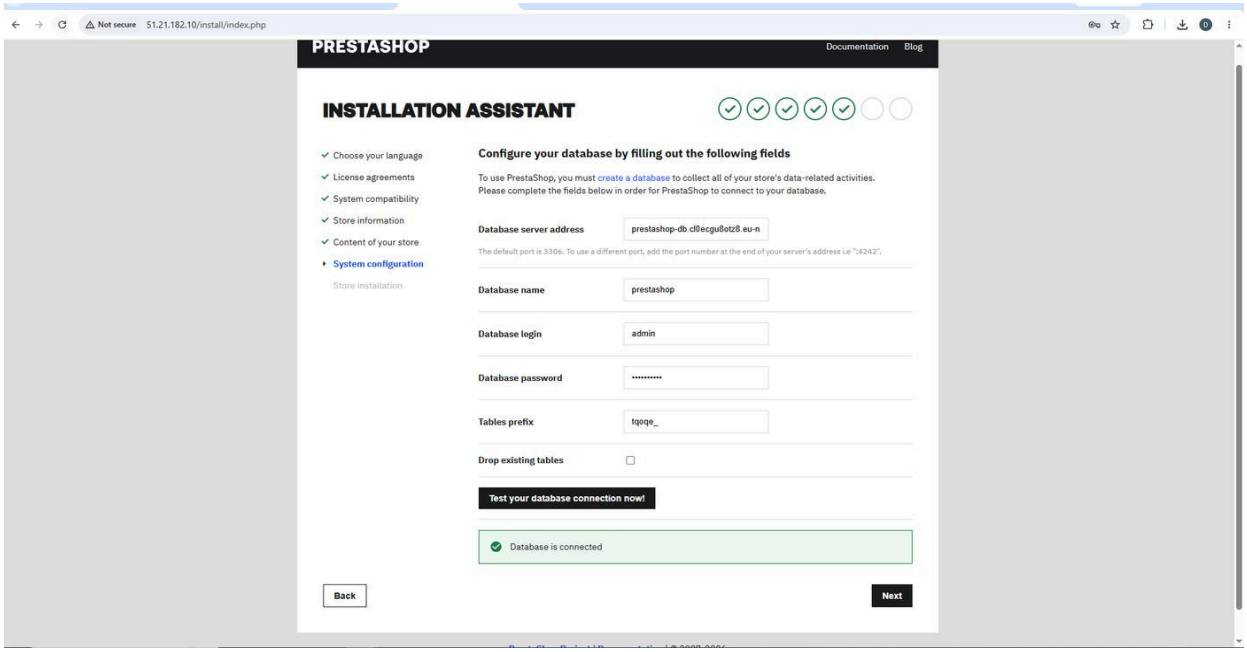


Figure 12: Successful database connection test from the PrestaShop installer to the remote RDS MySQL instance.

6.2 Installation Complete

The installer completed successfully, generating admin login credentials and a randomized, non-guessable admin folder path for security.

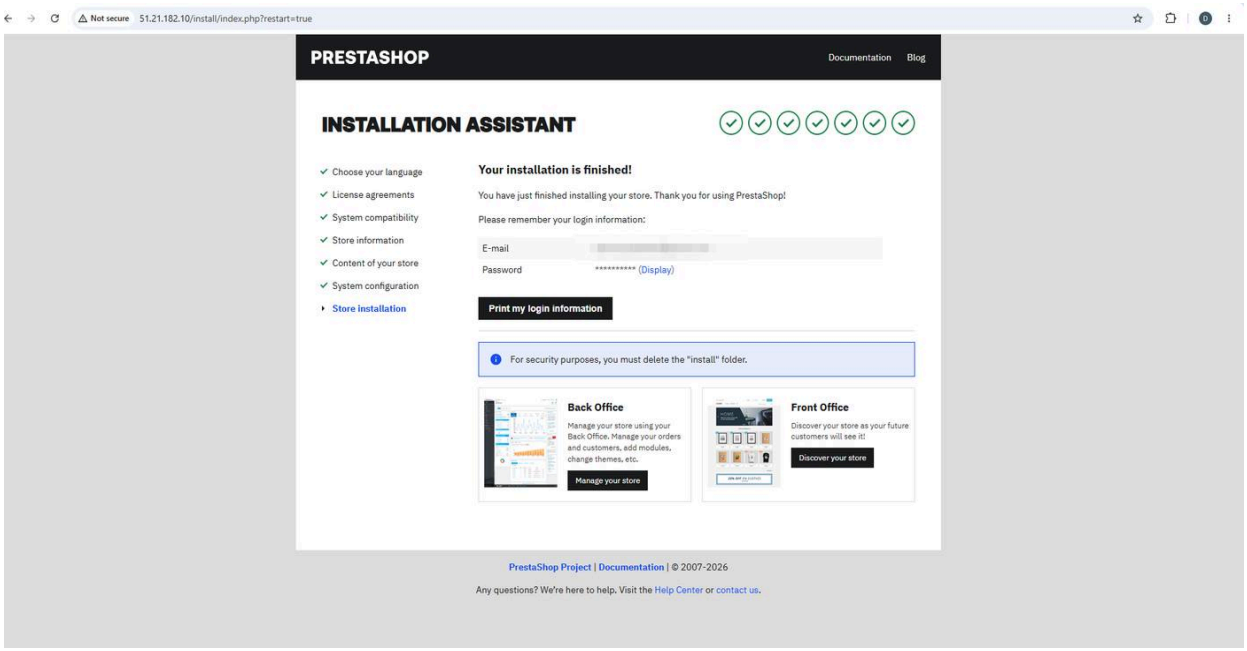


Figure 13: PrestaShop installation completed successfully.

As instructed by the installer, the /install directory was deleted from the server immediately afterward for security, since PrestaShop restricts full site access until this is done.

7. Verifying the Live Installation

With the installation complete, both the public storefront (front office) and the administrative dashboard (back office) were verified to confirm the deployment was fully functional.

7.1 Public Storefront

The storefront is publicly accessible with no authentication required, fulfilling the assignment's core requirement.

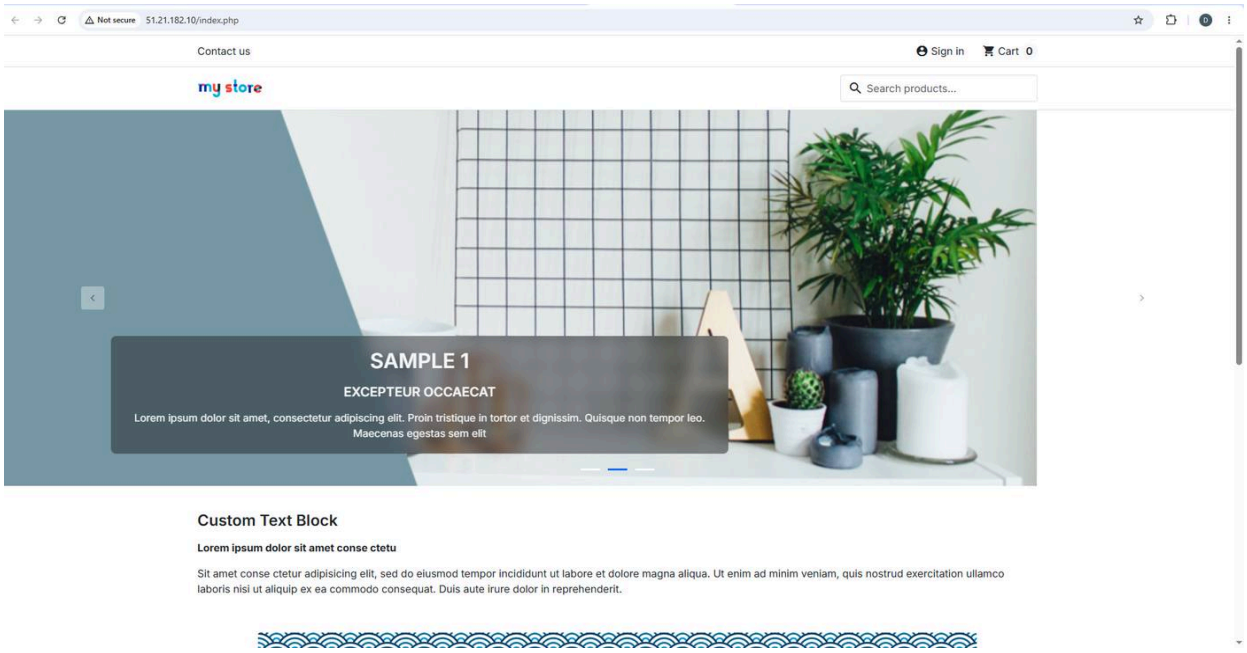


Figure 14: Live PrestaShop storefront accessible at http://51.21.182.10.

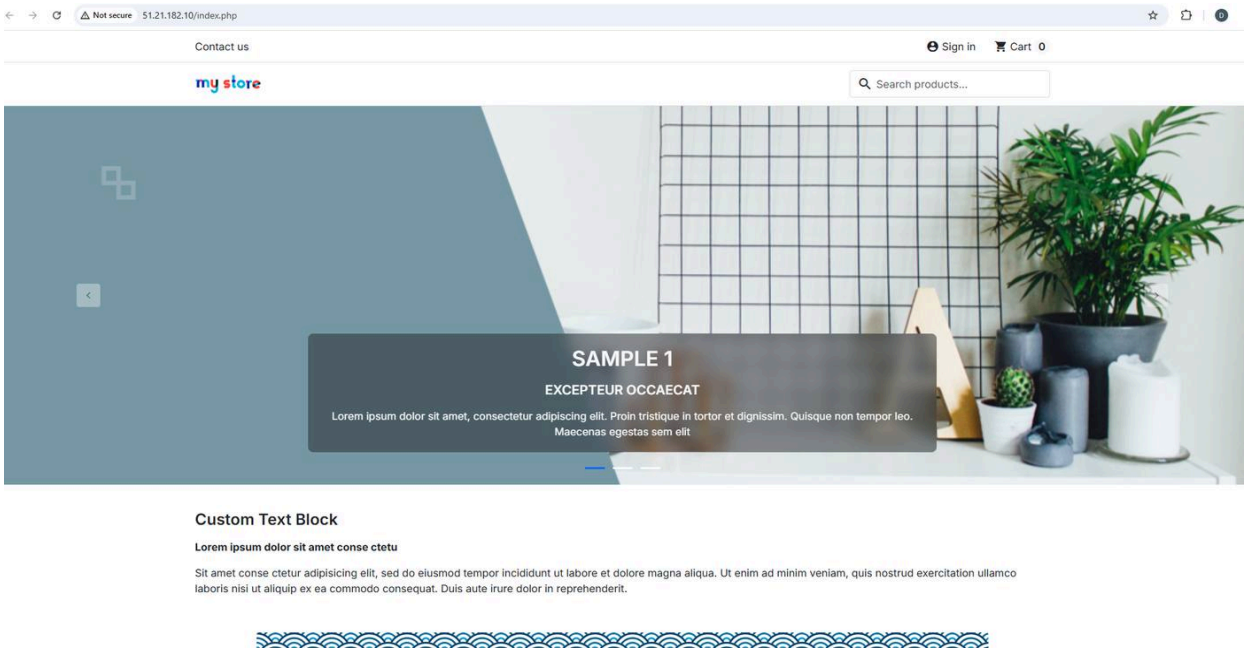


Figure 15: Storefront homepage loading correctly with the default Hummingbird theme.

7.2 Admin Back Office

The admin panel, located at a randomized URL path for security (e.g. /[hidden] /), was accessed using the credentials generated during installation.

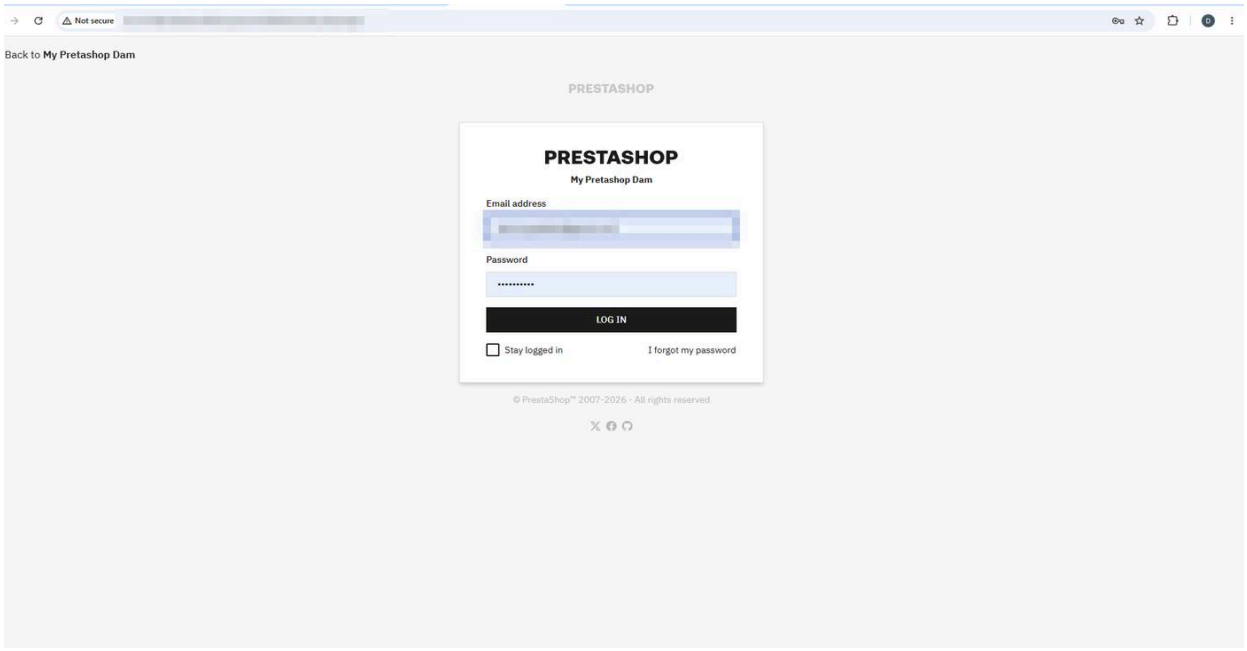


Figure 16: PrestaShop back office login screen.

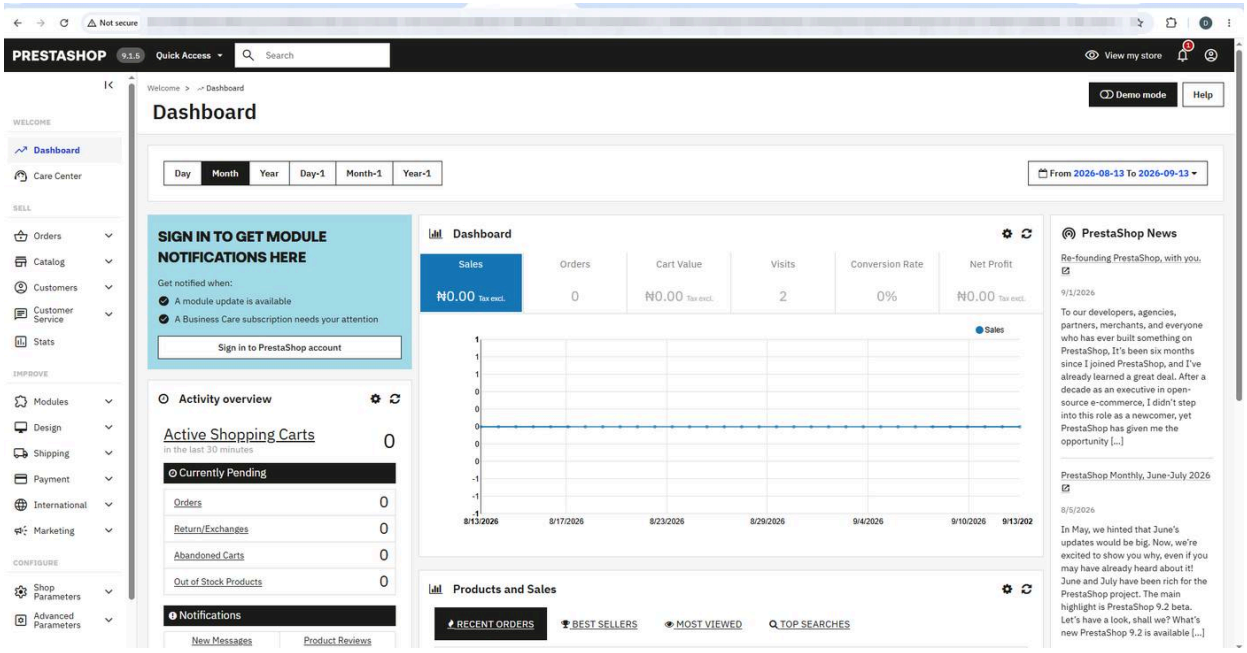


Figure 17: Successfully authenticated PrestaShop admin dashboard, confirming full administrative access.

8. Summary of Key Configuration Decisions

Database isolation

The RDS database was deployed as a separate resource from the EC2 application server, with Public Access set to No, so it has no direct exposure to the internet, satisfying the assignment's explicit requirement.

Least-privilege network access

Rather than allowing a broad IP range to reach the database, the RDS security group's inbound rule references the EC2 instance's security group directly, meaning only that specific application server can connect to the database on port 3306.

AWS Free Tier compliance

Both the EC2 instance (t3.micro) and RDS instance (db.t3.micro, Free Tier template) were selected specifically for Free Tier eligibility, keeping the deployment cost-free.

Public accessibility

The default public IPv4 address automatically assigned by AWS (51.21.182.10) was used as the store's URL, per the assignment's instruction to use the default address from the hosting provider rather than a custom domain.

Version selection

PrestaShop 9.1.5 was chosen over older 8.x releases because it is the first version line with official support for PHP 8.5, which ships by default on the latest Ubuntu LTS AMI available on EC2.

9. Conclusion

A fully functional PrestaShop 9.1.5 online store was successfully deployed on AWS Free Tier resources, with the database isolated on a separate RDS instance as required. The final store is publicly reachable at:

http://51.21.182.10